Contracts to Effects Compiler Correctness

(Deep Handler)

Overview

Proof for a compiler from Contracts to Effects using the deep handler, where the continuation automatically reinstalls the handler.

$$\mathtt{h}_{\mathscr{C}} : \mathtt{lab} \quad = \quad \{\mathscr{C} \to \forall\alpha.\langle \mathtt{blab}, \langle \alpha \to \mathsf{bool}, \alpha\rangle\rangle \to \alpha\}$$

$$\mathsf{wrap} \quad = \quad \mu w : \forall\alpha.\big(\big(() \to^{\langle \mathtt{h}_{\mathscr{C}}\rangle} \alpha\big) \to \alpha\big).\Lambda\alpha.\ \lambda f : \big(() \to^{\langle \mathtt{h}_{\mathscr{C}}\rangle} \alpha\big).\mathtt{handler}_d(h)\ f$$

$$h \quad = \quad \{\mathscr{C} \to \lambda\langle l, \langle e, v\rangle\rangle.\lambda k.\ \text{if } w[\mathsf{bool}]\ (\lambda_.e\ v) \text{ then } k\ v \text{ else } \mathtt{error}_l\}$$

where $\lambda\langle l, \langle e, v\rangle\rangle$ deconstructs the argument as tuple pattern

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Relations

Before we define relations that will be used in the simulation proof below. We define:

- Context E^* is any context without $\mathsf{check}^k_j(e, v)$ in Contracts.
- Context $E^+ = \Box \mid E^+\left[E^*\left[\mathsf{check}^k_j(\Box, v)\right]\right]$

$R_v \subseteq \text{Contracts } v \times \mathtt{Effects}\ v$

Abuse of notation: If $(v, v') \in R_v$, we write $v \sim_{R_v} v'$ or $v \sim v'$. We write $\tilde{v}$ for any value $\tilde{v}$ satisfying $v \sim \tilde{v}$.

$$\begin{aligned}
x &\sim x \\
b &\sim b \\
N &\sim N \\
\lambda x : \tau.e &\sim \lambda x : \tau^*.\tilde{e} \\
\mathsf{guard}^{k,l}_j(\kappa_1 \to \kappa_2, v) &\sim \tilde{e} \\
&\quad e = \lambda x : \tau.\mathsf{mon}^{k,l}_j\Big(\kappa_2, v\ \mathsf{mon}^{l,k}_j(\kappa_1, x)\Big) \\
\mathsf{guard}^{k,l}_j(\kappa_1 \to_i \lambda y.\kappa_2, v) &\sim \tilde{e} \\
&\quad e = \lambda x : \tau.\mathsf{mon}^{k,l}_j\Big(\kappa_3, v\ \mathsf{mon}^{l,k}_j(\kappa_1, x)\Big) \\
&\quad \kappa_3 = \kappa_2\left[y := \mathsf{mon}^{l,j}_j(\kappa_1, x)\right]
\end{aligned}$$

$R_e \subseteq \text{Contracts } e \times \mathtt{Effects}\ e$

Abuse of notation: If $(e, e') \in R_e$, we write $e \sim_{R_e} e'$ or $e \sim e'$. We write $\tilde{e}$ for any expression $\tilde{e}$ satisfying $e \sim \tilde{e}$.

$$
\begin{array}{rcl}
v & \sim & \tilde{v} \\
e_1\ e_2 & \sim & \tilde{e}_1\ \tilde{e}_2 \\
\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) & \sim & \texttt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle \rangle \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e) & \sim & \tilde{e}_1\ \tilde{e} \\
 & & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e) & \sim & \tilde{e}_1\ \tilde{e} \\
 & & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\
 & & \kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right] \\
\mathsf{blame}_j^k & \sim & \texttt{error}_k
\end{array}
$$

$R_{E^*} \subseteq \mathsf{Contracts}\ E^* \times \texttt{Effects}\ E^*$

Abuse of notation: If $(E^*, E'^*) \in R_{E^*}$, we write $E^* \sim_{R_{E^*}} E'^*$ or $E^* \sim E'^*$. We write $\tilde{E}^*$ for any expression $\tilde{E}^*$ satisfying $E^* \sim \tilde{E}^*$.

$$
\begin{array}{rcl}
\square & \sim & \square \\
E^*\ e & \sim & \tilde{E}^*\ \tilde{e} \\
v\ E^* & \sim & \tilde{v}\ \tilde{E}^* \\
\mathsf{mon}_j^{k,l}(\mathsf{flat}(e), E^*) & \sim & \texttt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{E}^* \rangle \rangle \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, E^*) & \sim & \tilde{e}_1\ \tilde{E}^* \\
 & & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, E^*) & \sim & \tilde{e}_1\ \tilde{E}^* \\
 & & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\
 & & \kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]
\end{array}
$$

$R_{E^+} \subseteq \mathsf{Contracts}\ E^+ \times \texttt{Effects}\ E^+$

Abuse of notation: If $(E^+, E'^+) \in R_{E^+}$, we write $E^+ \sim_{R_{E^+}} E'^+$ or $E^+ \sim E'^+$. We write $\tilde{E}^+$ for any expression $\tilde{E}^+$ satisfying $E^+ \sim \tilde{E}^+$.

$$\begin{array}{rcl} \square & \sim & \square \\ E^{+}\left[E^{*}\left[\mathsf{check}_{j}^{k}(\square, v)\right]\right] & \sim & \tilde{E}^{+}\left[\text{if } \square \text{ then } \left(\lambda x.\mathtt{handle}_{d}\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^{*}[x]\right)\right) \tilde{v} \text{ else } \mathtt{error}_{k}\right] \end{array}$$

Lemma 0.1. (Relation is closed) If $e \sim e'$ and $E^{*} \sim E'^{*}$ then $E^{*}[e] \sim E'^{*}[e']$.

Proof. By induction on the structure of $E^{*} \sim E'^{*}$.

- Case $\square \sim \square$

  $E^{*}[e] = e$ — Context Plug in
  $E'^{*}[e'] = e'$ — Context Plug in
  $e \sim e'$ — Given
  $E^{*}[e] \sim E'^{*}[e']$

- Case $E_1^{*}\ e_1 \sim \tilde{E}_1^{*}\ \tilde{e}_1$

  $E_1^{*} \sim \tilde{E}_1^{*}$ — Given
  $E_1^{*}[e] \sim \tilde{E}_1^{*}[e']$ — I.H.
  $E_1^{*}[e]\ e_1 \sim \tilde{E}_1^{*}[e']\ \tilde{e}_1$ — Relation
  $E^{*}[e] \sim E'^{*}[e']$

- Case $\mathsf{mon}_{j}^{k,l}(\mathsf{flat}(e_1), E_1^{*}) \sim \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{E}_1^{*}\rangle\rangle$

  $E_1^{*} \sim \tilde{E}_1^{*}$ — Given
  $E_1^{*}[e] \sim \tilde{E}_1^{*}[e']$ — I.H.
  $\mathsf{mon}_{j}^{k,l}(\mathsf{flat}(e_1), E_1^{*}[e]) \sim \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{E}_1^{*}[e']\rangle\rangle$ — Relation
  $E^{*}[e] \sim E'^{*}[e']$

- Case $\mathsf{mon}_{j}^{k,l}\left(\kappa_1 \to \kappa_2, \tilde{E}_1\right) \sim \tilde{e}_1\ \tilde{E}_1^{*}$, with $e_1 = \lambda f.\lambda x.\mathsf{mon}_{j}^{k,l}\left(\kappa_2, f\ \mathsf{mon}_{j}^{l,k}(\kappa_1, x)\right)$

  $e_1 \sim \tilde{e}_1$ — Given
  $E_1^{*} \sim \tilde{E}_1^{*}$ — Given
  $E_1^{*}[e] \sim \tilde{E}_1^{*}[e']$ — I.H.
  $\mathsf{mon}_{j}^{k,l}(\kappa_1 \to \kappa_2, E_1^{*}[e]) \sim \tilde{e}_1\ \tilde{E}_1^{*}[e']$ — Relation
  $E^{*}[e] \sim E'^{*}[e']$

- Case $\mathsf{mon}_{j}^{k,l}\left(\kappa_1 \to_i \lambda y.\kappa_2, \tilde{E}_1\right) \sim \tilde{e}_1\ \tilde{E}_1^{*}$, with $e_1 = \lambda f.\lambda x.\mathsf{mon}_{j}^{k,l}\left(\kappa_3, f\ \mathsf{mon}_{j}^{l,k}(\kappa_1, x)\right)$ and $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_{j}^{l,j}(\kappa_1, x)\right]$

  Similarly to the above case

- Other cases follow similarly.

□

Lemma 0.2. (Relation decomposition through check-free contexts)

If $E^*[e] \sim e'$, then there exist E'^* and e'_0 such that $E^* \sim E'^*$ and $e \sim e'_0$ and $e' = E'^*[e'_0]$.

Proof. By induction on the check-free context E^*.

- Case $E^* = \square$

$E^*[e] = e \sim e'$	Given
$E'^* = \square$	Choose
$e'_0 = e'$	Choose
$E^* \sim E'^*$	Definition
$e \sim e'_0$	Given
$e' = E'^*[e'_0]$	Context Plug in

- Case $E^* = E^*_0\ e_1$

$E^*_0[e]\ e_1 \sim e'$	Given
$E^*_0[e] \sim e'_0$	Relation inversion
$e_1 \sim e'_1$	Relation inversion
$e' = e'_0\ e'_1$	Relation inversion
$E^*_0 \sim E'^*_0$	I.H.
$e \sim e'_2$	I.H.
$e'_0 = E'^*_0[e'_2]$	I.H.
$E'^* = E'^*_0\ e'_1$	Choose
$E^* \sim E'^*$	Definition
$e' = E'^*[e'_2]$	Context Plug in

- Case $E^* = v\ E^*_0$

$v\ E^*_0[e] \sim e'$	Given
$v \sim v'$	Relation inversion
$E^*_0[e] \sim e'_0$	Relation inversion
$e' = v'\ e'_0$	Relation inversion
$E^*_0 \sim E'^*_0$	I.H.
$e \sim e'_1$	I.H.
$e'_0 = E'^*_0[e'_1]$	I.H.
$E'^* = v'\ E'^*_0$	Choose
$E^* \sim E'^*$	Definition
$e' = E'^*[e'_1]$	Context Plug in

- Case $E^* = \mathsf{mon}^{k,l}_j(\mathsf{flat}(e_0), E^*_0)$

$\mathsf{mon}^{k,l}_j(\mathsf{flat}(e_0), E^*_0[e]) \sim e'$	Given

$e_0 \sim e_0'$	Relation inversion
$E_0^*[e] \sim e_1'$	Relation inversion
$e' = \texttt{perform}\ \mathscr{C}\ \langle k, \langle e_0', e_1' \rangle \rangle$	Relation inversion
$E_0^* \sim E_0'^*$	I.H.
$e \sim e_2'$	I.H.
$e_1' = E_0'^*[e_2']$	I.H.
$E'^* = \texttt{perform}\ \mathscr{C}\ \langle k, \langle e_0', E_0'^* \rangle \rangle$	Choose
$E^* \sim E'^*$	Definition
$e' = E'^*[e_2']$	Context Plug in

- Case $E^* = \mathsf{mon}_j^{k,l}(\kappa, E_0^*)$

$\mathsf{mon}_j^{k,l}(\kappa, E_0^*[e]) \sim e'$	Given
$E_0^*[e] \sim e_0'$	Relation inversion
$e' = e_c'\ e_0'$	Relation inversion
$E_0^* \sim E_0'^*$	I.H.
$e \sim e_1'$	I.H.
$e_0' = E_0'^*[e_1']$	I.H.
$E'^* = e_c'\ E_0'^*$	Choose
$E^* \sim E'^*$	Definition
$e' = E'^*[e_1']$	Context Plug in

- Other cases follow similarly.

□

Lemma 0.3. (Relation is closed under substitution)

If $e \sim e'$ and $v \sim v'$ then $e[x := v] \sim e'[x := v']$.

Proof. By induction on the derivation of $e \sim e'$.

- Case $x \sim x$

$x[x := v] = v$	Replacement
$x[x := v'] = v'$	Replacement
$v \sim v'$	Given
$x[x := v] \sim x[x := v']$	

- Case $z \sim z$, where $z \neq x$

$z[x := v] = z$	Replacement
$z[x := v'] = z$	Replacement
$z[x := v] \sim z[x := v']$	Definition

- Case $\lambda z.e_1 \sim \lambda z.e_1'$

  $e_1[x := v] \sim e_1'[x := v']$ — I.H.
  $\lambda z.e_1[x := v] \sim \lambda z.e_1'[x := v']$ — Definition
  $(\lambda z.e_1)[x := v] \sim (\lambda z.e_1')[x := v']$ — Replacement

- Case $e_1\ e_2 \sim e_1'\ e_2'$

  $e_1[x := v] \sim e_1'[x := v']$ — I.H.
  $e_2[x := v] \sim e_2'[x := v']$ — I.H.
  $e_1\ e_2[x := v] = e_1[x := v]\ e_2[x := v]$ — Replacement
  $e_1'\ e_2'[x := v'] = e_1'[x := v']\ e_2'[x := v']$ — Replacement
  $e_1\ e_2[x := v] \sim e_1'\ e_2'[x := v']$ — Definition

- Case $\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) \sim \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle\rangle$

  $e_1[x := v] \sim \tilde{e}_1[x := v']$ — I.H.
  $e_2[x := v] \sim \tilde{e}_2[x := v']$ — I.H.
  $\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2)[x := v] \sim \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle\rangle[x := v']$ — Definition

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1) \sim \tilde{e}_2\ \tilde{e}_1$, with $e_2 = \lambda f.\lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, y)\right)$

  $e_1 \sim \tilde{e}_1$ — Given
  $e_2 \sim \tilde{e}_2$ — Given
  $e_1[x := v] \sim \tilde{e}_1[x := v']$ — I.H.
  $e_2[x := v] \sim \tilde{e}_2[x := v']$ — I.H.
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1)[x := v] = \mathsf{mon}_j^{k,l}(\kappa_1[x := v] \to \kappa_2[x := v], e_1[x := v])$
  $e_2[x := v] = \lambda f.\lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2[x := v], f\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], y)\right)$
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1)[x := v] \sim \tilde{e}_2[x := v']\ \tilde{e}_1[x := v']$ — Relation
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1)[x := v] \sim (\tilde{e}_2\ \tilde{e}_1)[x := v']$

  Choose variables, $f, y \neq x$ and $f, y \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, \tilde{e}_1) \sim \tilde{e}_2\ \tilde{e}_1$, with $e_2 = \lambda f.\lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, z)\right)$ and $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]$ and $y \neq x$

  $e_1 \sim \tilde{e}_1$ — Given
  $e_2 \sim \tilde{e}_2$ — Given
  $e_1[x := v] \sim \tilde{e}_1[x := v']$ — I.H.
  $e_2[x := v] \sim \tilde{e}_2[x := v']$ — I.H.
  $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_1)[x := v] =$ — Substitution
  $\quad \mathsf{mon}_j^{k,l}(\kappa_1[x := v] \to_i \lambda y.\kappa_2[x := v], e_1[x := v])$
  $e_2[x := v] = \lambda f.\lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3[x := v], f\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], z)\right)$

$\kappa_3[x := v] = \left(\kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]\right)[x := v]$
$\kappa_3[x := v] = (\kappa_2[x := v])\left[y := \mathsf{mon}_j^{l,j}(\kappa_1[x := v], z)\right]$ Substitution?
$\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_1)[x := v] \sim \tilde{e}_2[x := v']\ \tilde{e}_1[x := v']$ Relation
$\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_1)[x := v] \sim (\tilde{e}_2\ \tilde{e}_1)[x := v']$ Definition

Choose variables, $f, z \neq x$ and $f, y, z \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, \tilde{e}_1) \sim \tilde{e}_2\ \tilde{e}_1$, with $e_2 = \lambda f.\lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, z)\right)$ and $\kappa_3 = \kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]$

$e_1 \sim \tilde{e}_1$ Given
$e_2 \sim \tilde{e}_2$ Given
$e_1[x := v] \sim \tilde{e}_1[x := v']$ I.H.
$e_2[x := v] \sim \tilde{e}_2[x := v']$ I.H.
$\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, e_1)[x := v] =$ Substitution
$\quad \mathsf{mon}_j^{k,l}(\kappa_1[x := v] \to_i \lambda x.\kappa_2, e_1[x := v])$
$e_2[x := v] = \lambda f.\lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3[x := v], f\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], y)\right)$
$\kappa_3[x := v] = \left(\kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]\right)[x := v]$
$\kappa_3[x := v] = \left(\kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1[x := v], z)\right]\right)$
$\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, e_1)[x := v] \sim \tilde{e}_2[x := v']\ \tilde{e}_1[x := v']$ Relation
$\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, e_1)[x := v] \sim (\tilde{e}_2\ \tilde{e}_1)[x := v']$ Definition

Choose variables, $f, z \neq x$ and $f, y, z \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1) \sim \tilde{e}_1$, with $e_1 = \lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, y)\right)$

$e_1 \sim \tilde{e}_1$ Given
$e_1[x := v] \sim \tilde{e}_1[x := v']$ I.H.
$e_1[x := v] = \lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2[x := v], v_1[x := v]\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], y)\right)$
$\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)[x := v] = \mathsf{guard}_j^{k,l}(\kappa_1[x := v] \to \kappa_2[x := v], v_1[x := v])$
$\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)[x := v] \sim \tilde{e}_1[x := v']$ Relation

Choose variables, $y \neq x$ and $y \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) \sim \tilde{e}_1$, with $e_1 = \lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, z)\right)$ and $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right]$

$e_1 \sim \tilde{e}_1$ Given
$e_1[x := v] \sim \tilde{e}_1[x := v']$ I.H.
$e_1[x := v] = \lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_3[x := v], v_1[x := v]\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], x)\right)$
$\kappa_3[x := v] = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right][x := v]$
$\kappa_3[x := v] = (\kappa_2[x := v])\left[y := \mathsf{mon}_j^{l,j}(\kappa_1[x := v], z)\right]$ Substitution?

$\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)[x := v] =$

$\quad \mathsf{guard}_j^{k,l}(\kappa_1[x := v] \to_i \lambda y.\kappa_2[x := v], v_1[x := v])$

$\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)[x := v] \sim \tilde{e}[x := v']$ Relation

Choose variables, $z \neq x$ and $y, z \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Case $\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, v) \sim \tilde{e}_1$, with $e_1 = \lambda z.\mathsf{mon}_j^{k,l}\left(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$ and $\kappa_3 = \kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$

$e_1 \sim \tilde{e}_1$ Given

$e_1[x := v] \sim \tilde{e}_1[x := v']$ I.H.

$e_1[x := v] = \lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_3[x := v], v_1[x := v]\ \mathsf{mon}_j^{l,k}(\kappa_1[x := v], x)\right)$

$\kappa_3[x := v] = \kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1, z)\right][x := v]$

$\kappa_3[x := v] = \kappa_2\left[x := \mathsf{mon}_j^{l,j}(\kappa_1[x := v], z)\right]$

$\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, v_1)[x := v] =$

$\quad \mathsf{guard}_j^{k,l}(\kappa_1[x := v] \to_i \lambda x.\kappa_2, v_1[x := v])$

$\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda x.\kappa_2, v_1)[x := v] \sim \tilde{e}[x := v']$ Relation

Choose variables, $z \neq x$ and $z \notin \mathrm{fv}(v, v')$ by alpha-renaming.

- Other cases follows similarly.

□

Lemma 0.4. (Breaking down Context E)

For all contexts E, uniquely decompose $E = E^+[E^*]$.

Uniqueness observation: Recall that E^* has no check contexts.

- If E does not have check, then it is $E^* = E$, and $E^+ = \square$.
- Else there is only a single decomposition to get out E^* without check contexts

Proof. By induction on E.

- Case $E = \square$

Choose $E^+ = E^* = \square$

- Case $E = E_1\ e$

$E_1 = E_1^+[E_1^*]$ I.H.

$E = E_1^+[E_1^*]\ e$

$E = (E_1^+\ e)[E_1^*]$

▸ Case $E_1^+ = \square$

$E = E_1^*\ e$

$E^+ = \square$ Choose
$E^* = E_1^*\ e$ Choose

- Case $E_1^+ \neq \square$

  $E^+ = E_1^+\ e$ Lemma 0.5 ($K = \square\ e$) and Choose
  $E^* = E_1^*$ Choose

- Case $E = \mathsf{check}_j^k(E_1, v)$

  $E_1 = E_1^+[E_1^*]$ I.H.
  $E = \mathsf{check}_j^k(E_1^+[E_1^*], v) = \mathsf{check}_j^k(E_1^+, v)[E_1^*]$

  - Case $E_1^+ = \square$

    $E = \mathsf{check}_j^k(\square, v)[E_1^*]$
    $E^+ = \mathsf{check}_j^k(\square, v), E^* = E_1^*$ Choose

  - Case $E_1^+ \neq \square$

    $E_2^+ = \mathsf{check}_j^k(E_1^+, v)$ Lemma 0.5 ($K = \mathsf{check}_j^k(\square, v)$)
    $E = E_2^+[E^*]$

□

Lemma 0.5.

If $E_1^+ \neq \square$ then there is a E_2^+, $E_2^+ = K[E_1^+]$.

Where $K = \square\ e \mid v\ \square \mid \mathsf{mon}_j^{k,l}(\kappa, \square) \mid \mathsf{check}_j^k(\square, v)$

Proof.

By induction on the structure of E_1^+.

- Case $\square$ invalid.
- Case $E_3^+\left[E^*\left[\mathsf{check}_j^k(\square, v)\right]\right]$

  - Case $E_3^+ = \square$

    – Case $K = \mathsf{check}_{j_1}^{k_1}(\square, v_1)$

      $K = E_4^+$ Definition of E^+
      $K[E_1^+] = E_4^+\left[E^*\left[\mathsf{check}_j^k(\square, v)\right]\right]$
      $E_2^+ = E_4^+\left[E^*\left[\mathsf{check}_j^k(\square, v)\right]\right]$ Choose

    – Case $K \neq \mathsf{check}_{j_1}^{k_1}(\square, v_1)$

      $K = E_4^*$ Definition of E^*
      $K[E_1^+] = E_4^*\left[E^*\left[\mathsf{check}_j^k(\square, v)\right]\right] = E_5^*\left[\mathsf{check}_j^k(\square, v)\right]$
      $E_2^+ = \square\left[E_5^*\left[\mathsf{check}_j^k(\square, v)\right]\right]$ Choose

- Case $E_3^+ \neq \square$

  $E_4^+ = K[E_3^+]$ I.H.
  $K[E_1^+] = K[E_3^+[E^*[\mathsf{check}_j^k(\square, v)]]] =$

  $K[E_3^+][E^*[\mathsf{check}_j^k(\square, v)]] = E_4^+[E^*[\mathsf{check}_j^k(\square, v)]]$

  $E_2^+ = E_4^+[E^*[\mathsf{check}_j^k(\square, v)]]$ Choose

  $\square$

Lemma 0.6. (Compiler is subset of relation)

If $\Gamma \vdash e_1 : \tau \rightarrowtail^d e_2$ then $e_1 \sim e_2^*$.

In particular, for closed programs, if $\cdot \vdash e_1 : \tau \rightarrowtail^d e_2$ then $e_1 \sim e_2^*$.

In this lemma, we use e^* as shorthand to denote e untyped.

Proof.

By induction on structure of $\Gamma \vdash e_1 : \tau \rightarrowtail^d e_2$.

- Case

$$\frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau \rightarrowtail^d x}$$

$e_2^* = x$ Definition
$x \sim x$ Relation

- Case

$$\frac{\Gamma, x : \tau_1 \vdash e : \tau_2 \rightarrowtail^d e'}{\Gamma \vdash \lambda x : \tau_1.e : \tau_1 \to \tau_2 \rightarrowtail^d \lambda x : \tau_1.e'}$$

$e_2^* = \lambda x.e'^*$ Definition
$e \sim e'^*$ I.H. on the premise with context $\Gamma, x : \tau_1$
$\lambda x.e \sim \lambda x.e'^*$ Relation

- Case

$$\frac{\Gamma \vdash e_3 : \tau \to \mathsf{bool} \rightarrowtail^d e_5 \qquad \Gamma \vdash e_4 : \tau \rightarrowtail^d e_6}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_3), e_4) : \tau \rightarrowtail^d \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_5, e_6 \rangle\rangle}$$

$e_2^* = \mathtt{perform}\ \mathscr{C}\ \langle k, \langle e_5^*, e_6^* \rangle\rangle$ Definition
$e_3 \sim e_5^*$ I.H.
$e_4 \sim e_6^*$ I.H.
$e_1 \sim e_2^*$ Relation

- Case

$$\frac{\Gamma \vdash e_3 : \tau_2 \rightarrowtail^d e_4 \qquad \Gamma \vdash \lambda f : \tau_1 \to \tau_2 . \lambda x : \tau_1 . \mathsf{mon}_j^{k,l}\big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail^d e_5}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail^d e_5\ e_4}$$

$e_3 \sim e_4^*$ I.H.
$\lambda f : \tau_1 \to \tau_2 . \lambda x : \tau_1 . \mathsf{mon}_j^{k,l}\big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big) \sim e_5^*$ I.H.
$e_1 \sim e_2^*$ Relation

- Case

$$\frac{\Gamma \vdash e_3 : \tau_1 \to \tau_2 \rightarrowtail^d e_4 \qquad \Gamma \vdash \lambda f . \lambda x . \mathsf{mon}_j^{k,l}\big(\kappa_2\big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\big], f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail^d e_5}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y . \kappa_2, e_3) : \tau_1 \to \tau_2 \rightarrowtail^d e_5\ e_4}$$

$e_3 \sim e_4^*$ I.H.
$\lambda f . \lambda x . \mathsf{mon}_j^{k,l}\big(\kappa_2\big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\big], f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big) \sim e_5^*$ I.H.
$e_1 \sim e_2^*$ Relation

- Other cases are similar.

□

Compiler Correctness

Figure 1 shows the structure of the proof. The source program e compiles to e', and the target program runs e' under wrap. After the handler is installed, the target reaches a state e_0'' with $\mathrm{EQ}(e, e_0'')$ (Lemma 0.8). From then on, every source step $e_i \twoheadrightarrow e_{i+1}$ is matched by zero or more target steps $e_i'' \Longrightarrow^* e_{i+1}''$ that keep $\mathrm{EQ}(e_{i+1}, e_{i+1}'')$ (Lemma 0.10). Chaining the squares (Lemma 0.9), the source result r and the target result $\tilde{r}$ are related: a value goes to its related value, and blame_j^k goes to $\mathtt{error}_k$.

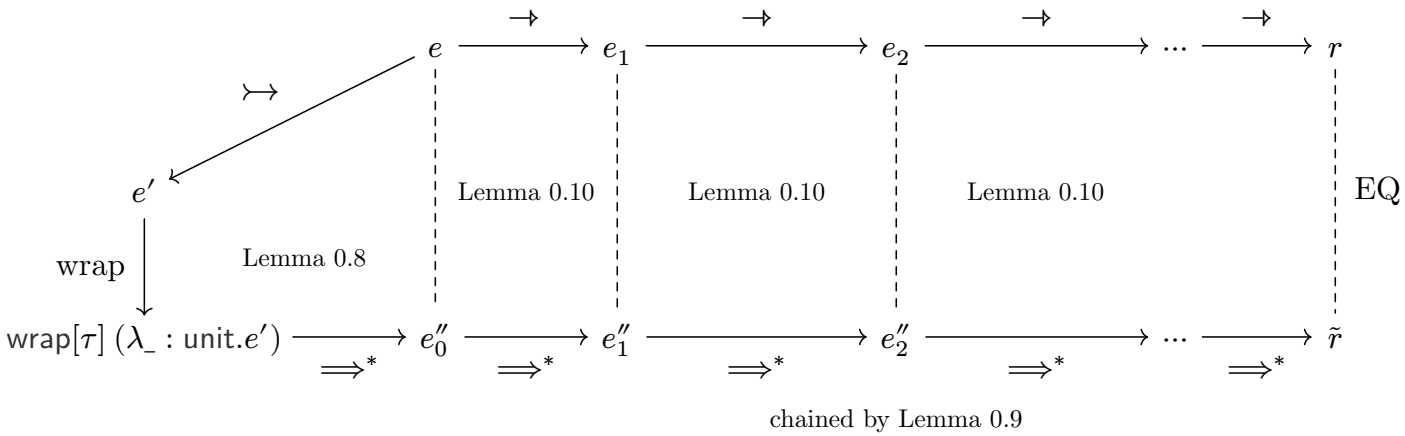


Figure 1: Structure of the compiler correctness proof. Dashed lines rel states by EQ.

Define a predicate between Contracts e and Untyped Effects e'.

$$\mathrm{EQ}(e, e') = \begin{cases} \text{true if } e = v \text{ and } e' = \tilde{v} \\ \text{true if } e' = \mathtt{handle}_d(\mathtt{h}_{\mathscr{C}}, \tilde{e}) \text{ and } e \neq E^*\left[\mathsf{check}^k_j(e'', v)\right] \\ \text{true if } e = E^+\left[E^*\left[\mathsf{check}^k_j(e_1, v)\right]\right] \text{ and} \\ \qquad e' = \tilde{E}^+\left[\text{if } e'_1 \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[x]\right)\right) \tilde{v} \text{ else } \mathtt{error}_j\right] \text{ and} \\ \qquad \mathrm{EQ}(e_1, e'_1) \\ \text{true if } e = \mathsf{blame}^k_j \text{ and } e' = \mathtt{error}_k \\ \text{false otherwise} \end{cases}$$

Theorem 0.7. (Compiler Correctness)

If $\cdot \vdash e : \tau \rightarrowtail^d e'$ and $e^* = \mathrm{untyped}(\mathsf{wrap}[\tau]\ (\lambda_ : \mathsf{unit}.e'))$ and $e \twoheadrightarrow^* b$ then $e^* \Longrightarrow^* b$.

Proof.

$e^* \Longrightarrow^* e'_1$	Lemma 0.8
$\mathrm{EQ}(e, e'_1)$	Lemma 0.8
$e'_1 \Longrightarrow^* b$	Lemma 0.9
$e^* \Longrightarrow^* b$	Combining steps

□

Lemma 0.8. (Wrapped program steps into an EQ state with source)

If $\cdot \vdash e : \tau \rightarrowtail^d e'$ and $e^*_1 = \mathsf{wrap}[\tau]\ (\lambda_ : \mathsf{unit}.e')$ then it exists e^*_2 such that $e^*_1 \Longrightarrow^* e^*_2$ and $\mathrm{EQ}(e, e^*_2)$.

Proof.

$e \sim e'^*$	Lemma 0.6
$e^*_1 \Longrightarrow^* \mathtt{handler}_d(\mathtt{h}_{\mathscr{C}})\ \lambda_.e'^* \Longrightarrow^* \mathtt{handle}_d(\mathtt{h}_{\mathscr{C}}, e'^*) = e^*_2$	

$\mathrm{EQ}(e, e_2^*)$ Definition

□

Lemma 0.9. If $e \twoheadrightarrow^* r$ and $\mathrm{EQ}(e, e')$ then $e' \Longrightarrow^* \tilde{r}$.

Proof. By breaking down the $\twoheadrightarrow^*$, and induction on it.

- Case $v \twoheadrightarrow^* v$

  $e' = \tilde{r}$ Definition of $\mathrm{EQ}(e, e')$
  $e' \Longrightarrow^* \tilde{r}$ Reflexivity

- Case $\mathsf{blame}_j^k \twoheadrightarrow^* \mathsf{blame}_j^k$

  $r = \mathsf{blame}_j^k$ Case assumption
  $\tilde{r} = \mathtt{error}_j$ Relation
  $e' = \mathtt{error}_j = \tilde{r}$ Definition of $\mathrm{EQ}(e, e')$
  $e' \Longrightarrow^* \tilde{r}$ Reflexivity

- Case $e \twoheadrightarrow e_1$ and $e_1 \twoheadrightarrow^* r$

  $e' \Longrightarrow^* e_1'$ Lemma 0.10
  $\mathrm{EQ}(e_1, e_1')$ Lemma 0.10
  $e_1' \Longrightarrow^* \tilde{r}$ I.H.
  $e' \Longrightarrow^* \tilde{r}$ Chaining steps

□

Lemma 0.10. If $e_1 \twoheadrightarrow e_2$ and $\mathrm{EQ}(e_1, e_1')$ then there exists e_2' such that $e_1' \Longrightarrow^* e_2'$ and $\mathrm{EQ}(e_2, e_2')$.

Proof of Lemma 0.10. By case analysis on $e_1 \twoheadrightarrow e_2$ with respect to $\mathrm{EQ}(e_1, e_1')$.

- Case $E\left[\mathsf{blame}_j^k\right] \twoheadrightarrow \mathsf{blame}_j^k$

  $E = E^+[E^*]$ Lemma 0.4

  - Case $E^+ = \square$

    $E^*\left[\mathsf{blame}_j^k\right] \sim \tilde{E}^*[\mathtt{error}_k]$ Lemma 0.2
    $\tilde{E}^*[\mathtt{error}_k] \Longrightarrow \mathtt{error}_k$
    $\mathsf{blame}_j^k \sim \mathtt{error}_k$ Definition
    $\mathrm{EQ}\left(\mathsf{blame}_j^k, \mathtt{error}_k\right)$ EQ Definition
    $e_2' = \mathtt{error}_k$ Choose

  - Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

    $E\left[\mathsf{blame}_j^k\right] = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(\mathsf{blame}_j^k, v_1\right)\right]\right]$

$\mathsf{blame}_j^k \sim \mathtt{error}_k$ Definition
$e_1 \sim \tilde{E}_1^+\left[\text{if } \mathtt{error}_k \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}_1^*[x]\right)\right) \tilde{v} \text{ else } \mathtt{error}_{k_1}\right]$
$e_1' \Longrightarrow \mathtt{error}_k$
$\mathrm{EQ}\left(\mathsf{blame}_j^k, \mathtt{error}_k\right)$ EQ Definition
$e_2' = \mathtt{error}_k$ Choose

- Case $E[e_3] \to E[e_4]$ where $e_3 \dashrightarrow e_4$

  Overview of proof strategy: breaking down $E = E^+[E^*]$ using Lemma 0.4 and case on E^+, then try to match $\mathrm{EQ}(e_1, e_1')$ cases. EQ case 2 holds if and only if $E^+ = \square$, and case 3 holds if and only if $E^+ \neq \square$ (other case) easily observed the form of e_1 in EQ. By step it should arrive at the next state that matches EQ predicate cases.

  During the reduction, we likely encounter $\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[e]\right)$. By Lemma 0.11, the handler invoked is obviously the current handler.

  ▸ Case $e_3 = \mathsf{mon}_j^{k,l}(\mathsf{flat}(e), v) \dashrightarrow \mathsf{check}_j^k(e\ v, v) = e_4$

  $E = E^+[E^*]$ Lemma 0.4
  $e_1 = E^+[E^*[e_3]]$ Substitution

  – Case $E^+ = \square$

  $E^* \sim \tilde{E}^*$ Lemma 0.2
  $e_3 \sim \tilde{e}_3$ Lemma 0.2
  $\tilde{e}_1 = \tilde{E}^*[\tilde{e}_3]$ Lemma 0.2
  $e_1 = E^*[e_3] \sim \tilde{E}^*[\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v}\rangle\rangle]$ Relation
  $e_1' = \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v}\rangle\rangle]\right)$ Case 2
  $e_1' \Longrightarrow^*$
  $\text{if } \mathtt{handle}_d(\mathtt{h}_{\mathscr{C}}, \tilde{e}\ \tilde{v}) \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[x]\right)\right) \tilde{v} \text{ else } \mathtt{error}_j$
  $e_2' =$ Choose
  $\text{if } \mathtt{handle}_d(\mathtt{h}_{\mathscr{C}}, \tilde{e}\ \tilde{v}) \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[x]\right)\right) \tilde{v} \text{ else } \mathtt{error}_j$
  $\mathrm{EQ}(e\ v, \mathtt{handle}_d(\mathtt{h}_{\mathscr{C}}, \tilde{e}\ \tilde{v}))$ Case 2
  $\mathrm{EQ}(e_2, e_2')$ Case 3

  – Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

  $e_1 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(E^*[e_3], v_1)\right]\right]$
  $e_3 \sim \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v}\rangle\rangle = \tilde{e}_3$
  $\mathrm{EQ}\left(E^*[e_3], \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\tilde{e}_3]\right)\right)$ Case 2, by shape
  $\mathrm{EQ}\left(e_1, \tilde{E}_1^+\left[\text{if } \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\tilde{e}_3]\right) \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}_1^*[x]\right)\right) \tilde{v}_1 \text{ else } \mathtt{error}_{j_1}\right]\right)$
  $e_1' = \tilde{E}_1^+\left[\text{if } \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\tilde{e}_3]\right) \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}_1^*[x]\right)\right) \tilde{v}_1 \text{ else } \mathtt{error}_{j_1}\right]$
  Identified e_1' based on the condition of EQ case 3
  $e_1' \Longrightarrow^* \tilde{E}_2^+\left[\text{if } \mathtt{handle}_d(\mathtt{h}_{\mathscr{C}}, \tilde{e}\ \tilde{v}) \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[x]\right)\right) \tilde{v} \text{ else } \mathtt{error}_j\right]$

$e_2' = \tilde{E}_2^+\left[\text{if } \mathtt{handle}_d(\mathtt{h}_{\mathscr{C}}, \tilde{e}\ \tilde{v}) \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[x]\right)\right)\ \tilde{v} \text{ else } \mathtt{error}_j\right]$ Choose

where $\tilde{E}_2^+ = \tilde{E}_1^+\left[\text{if } \square \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}_1^*[x]\right)\right)\ \tilde{v}_1 \text{ else } \mathtt{error}_{j_1}\right]$

$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{check}_j^k(e\ v, v)\right], v_1\right)\right]\right]$

$\mathrm{EQ}\left(E^*\left[\mathsf{check}_j^k(e\ v, v)\right], \text{if } \mathtt{handle}_d(\mathtt{h}_{\mathscr{C}}, \tilde{e}\ \tilde{v}) \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[x]\right)\right)\ \tilde{v} \text{ else } \mathtt{error}_j\right)$

$\mathrm{EQ}(e_2, e_2')$ Case 3

(Case 2 is invalid, because of the shape of e_2)

- Case $e_3 = \mathsf{check}_j^k(\mathsf{true}, v) \dashrightarrow v = e_4$

$E = E^+[E^*]$ Lemma 0.4

$e_1 = E^+[E^*[e_3]]$ Substitution

– Case $E^+ = \square$

Let $e_5 = \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[x]\right)\right)\ \tilde{v}$.

$e_1' = \text{if } e_3' \text{ then } e_5 \text{ else } \mathtt{error}_j$ $\mathrm{EQ}(e_1, e_1')$, case 3

$\mathrm{EQ}(\mathsf{true}, e_3')$ Since true is a value

$e_3' = \mathsf{true}$ Definition of EQ

$e_1' \Longrightarrow^* \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\tilde{v}]\right) = e_2'$

$e_2 = E^*[v]$

$\mathrm{EQ}(e_2, e_2')$ Case 2

– Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

Let $e_5 = \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[x]\right)\right)\ \tilde{v}$ and $e_6 = \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}_1^*[x]\right)\right)\ \tilde{v}_1$.

$e_1' =$ $\mathrm{EQ}(e_1, e_1')$, case 3

$\tilde{E}_1^+\left[\text{if if } e_3' \text{ then } e_5 \text{ else } \mathtt{error}_j \text{ then } e_6 \text{ else } \mathtt{error}_{j_1}\right]$

$\mathrm{EQ}(\mathsf{true}, e_3')$ Since true is a value

$e_3' = \mathsf{true}$ Definition of EQ

$e_1' \Longrightarrow^* \tilde{E}_1^+\left[\text{if } \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\tilde{v}]\right) \text{ then } e_6 \text{ else } \mathtt{error}_{j_1}\right] =$

e_2'

$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(E^*[v], v_1)\right]\right]$

$\mathrm{EQ}\left(E^*[v], \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\tilde{v}]\right)\right)$ Case 2

$\mathrm{EQ}(e_2, e_2')$ Case 3

(Case 2 is invalid, because of the shape of e_2.)

- Case $e_3 = \mathsf{check}_j^k(\mathsf{false}, v) \dashrightarrow \mathsf{blame}_j^k = e_4$

$E = E^+[E^*]$ Lemma 0.4

$e_1 = E^+[E^*[e_3]]$ Substitution

– Case $E^+ = \square$

$e_1 = E^*\left[\mathsf{check}_j^k(\mathsf{false}, v)\right]$
$e_1' =$

– Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

$e_1 = E^+\left[E^*\left[\mathsf{check}_j^k(\mathsf{false}, v)\right]\right]$
$e_1' =$ EQ(e_1, e_1')
$\tilde{E}^+\left[\text{if false then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[x]\right)\right)\tilde{v} \text{ else } \mathtt{error}_j\right]$
$e_1' \Longrightarrow^* \tilde{E}^+\left[\mathtt{error}_j\right] = e_2'$ Reduction
$e_2 = E^+\left[E^*\left[\mathsf{blame}_j^k\right]\right]$ Source reduction
$\tilde{E}^+\left[\mathtt{error}_j\right] \Longrightarrow \mathtt{error}_j$ **Step-E-Error**
$\mathsf{blame}_j^k \sim \mathtt{error}_j$ Relation
EQ(e_2, e_2') Definition

▸ Case $e_3 = \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, v) \dashrightarrow \mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v) = e_4$

$E = E^+[E^*]$ Lemma 0.4
$e_1 = E^+[E^*[e_3]]$ Substitution

– Case $E^+ = \square$

Let $e_5 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) = \lambda f.\lambda x.e_{5a}$

$E^* \sim \tilde{E}^*$ Lemma 0.2
$e_3 \sim \tilde{e}_3$ Lemma 0.2
$\tilde{e}_1 = \tilde{E}^*[\tilde{e}_3]$ Lemma 0.2
$e_1 = E^*[e_3] \sim \tilde{E}^*[(\lambda f.\lambda x.\tilde{e}_{5a})\ \tilde{v}]$ Relation
$e_1' = \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[(\lambda f.\lambda x.\tilde{e}_{5a})\ \tilde{v}]\right)$ EQ(e_1, e_1'), case 2
$e_1' \Longrightarrow^* \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\lambda x.\tilde{e}_{5a}[f := \tilde{v}]]\right) = e_2'$ Reduction
$e_2 = E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right]$
$e_6 = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$ Let
$e_6 = e_5[f := v]$ Substitution
$\tilde{e}_6 = \tilde{e}_5[f := \tilde{v}]$ Lemma 0.3
$\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v) \sim \tilde{e}_6 = \tilde{e}_5[f := \tilde{v}]$ Definition
$E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right] \sim \tilde{E}^*[\tilde{e}_5[f := \tilde{v}]]$ Lemma 0.1
EQ(e_2, e_2') Case 2

– Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

Let $e_5 = \lambda f.\lambda x.e_{5a}$, where $e_{5a} = \mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$.

Let $\tilde{E}_2^+ = \tilde{E}_1^+\left[\text{if } \square \text{ then } \left(\lambda x.\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}_1^*[x]\right)\right)\tilde{v}_1 \text{ else } \mathtt{error}_{j_1}\right]$

$e_1' = \tilde{E}_2^+\left[\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\tilde{e}_5\ \tilde{v}]\right)\right]$

$e_1' \Longrightarrow^* \tilde{E}_2^+\left[\mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\lambda x.\tilde{e}_{5a}[f := \tilde{v}]]\right)\right] = e_2'$
$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right], v_1\right)\right]\right]$
$e_6 = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$ Let
$e_6 = e_{5a}[f := v]$ Substitution
$\tilde{e}_6 = \tilde{e}_{5a}[f := \tilde{v}]$ Lemma 0.3
$\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v) \sim \tilde{e}_6 = \tilde{e}_{5a}[f := \tilde{v}]$ Definition
$E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right] \sim \tilde{E}^*[\tilde{e}_{5a}[f := \tilde{v}]]$ Lemma 0.1
$\mathrm{EQ}\left(E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right], \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\tilde{e}_{5a}[f := \tilde{v}]]\right)\right)$ Case 1
$\mathrm{EQ}(e_2, e_2')$ Case 3

- Case $e_3 = \mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}_j^{k,l}\left(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right) = e_4$

  $E = E^+[E^*]$ Lemma 0.4
  $e_1 = E^+[E^*[e_3]]$ Substitution

  – Case $E^+ = \square$

    Let $e_5 = \lambda f.\lambda x.e_{5a}$, where $e_{5a} = \mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$.

    $E^* \sim \tilde{E}^*$ Lemma 0.2
    $e_3 \sim \tilde{e}_3$ Lemma 0.2
    $\tilde{e}_1 = \tilde{E}^*[\tilde{e}_3]$ Lemma 0.2
    $\tilde{e}_3 = \tilde{e}_5\ \tilde{v}_2$ Relation
    $e_1' = \mathtt{handle}_d\left(\mathtt{h}_{\mathscr{C}}, \tilde{E}^*[\tilde{e}_3]\right)$ $\mathrm{EQ}(e_1, e_1')$, case 2
    $e_1' \Longrightarrow^* e_1' = e_2'$ Reflexivity
    $\tilde{e}_4 = \tilde{e}_5\ \tilde{v}_2$ Relation
    $E^*[e_4] \sim \tilde{E}^*[\tilde{e}_4]$ Lemma 0.1
    $\mathrm{EQ}(e_2, e_2')$ Case 2

  – Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

    Same as the case $E^+ = \square$, inside $\tilde{E}^+$, by case 3.

- Case $e_3 = \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) \dashrightarrow \mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) = e_4$

  Similar to its functional variant.

- Case $e_3 = \mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}_j^{k,l}\left(\kappa_3, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right) = e_4$ where $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, v_2)\right]$

  Similar to its functional variant.

□

Lemma 0.11. ($\tilde{E}^*$ does not have a handler)

If $E^* \sim \tilde{E}^*$, then $\mathscr{C} \notin \mathtt{bop}(\tilde{E}^*)$.

Proof.

By induction on the structure of $E^* \sim \tilde{E}^*$. Trivial, the related context has no handler.$\square$